📘 Computer Networks | Class 12 CS

Part 1: Evolution of Networking

1. Introduction to Computer Networks

> **Definition:** A **computer network** is a collection of interconnected computers and other devices (like printers, scanners) that can communicate with each other and share resources (like files, printers, and internet connection).

Key Advantages:

- **Resource Sharing:** Share hardware (printers), software, and data.
- **Communication:** Email, instant messaging, video conferencing.
- **Data Centralization:** Store data on a central server for easy access and backup.
- **Cost Efficiency:** Reduce costs by sharing expensive resources.
- **Reliability:** If one computer fails, others can take over its work.

2. Evolution of Networking

The evolution of networking can be traced through three key milestones:

a) ARPANET (Advanced Research Projects Agency Network) — 1969

- **What it was:** The world's first operational packet-switching network and the direct predecessor of the Internet.
- **Who created it:** Funded by the U.S. Department of Defense.
- **Purpose:** To create a communication network that could survive a partial outage (e.g., during a war). If one path was destroyed, data could be rerouted through another path.
- **Significance:** It introduced the concept of **packet switching** and the **TCP/IP** protocol suite.

b) NSFNET (National Science Foundation Network) — 1986

- **What it was:** A network created by the National Science Foundation (NSF) to connect supercomputing centers across the U.S. and provide access to researchers.
- **Purpose:** To promote scientific research and education.
- **Significance:** It was a major backbone of the early Internet and helped to popularize it among a wider audience beyond the military. It was eventually decommissioned in 1995, paving the way for commercial Internet service providers (ISPs).

c) INTERNET (INTERconnected NETworks)

- **What it is:** The global system of interconnected computer networks that use the Internet protocol suite (TCP/IP) to link devices worldwide. It is a "network of networks."
- **How it evolved:** The Internet grew out of ARPANET and NSFNET. The key development was the adoption of **TCP/IP** as the standard communication protocol on January 1, 1983. The invention of the **World Wide Web (WWW)** by Tim Berners-Lee in 1989 made the Internet easily accessible and popular among the general public.

Part 2: Data Communication Terminologies

1. Concept of Communication

> **Communication** is the process of exchanging information between two or more entities. In the context of computers, it is the transfer of data from one point to another.

2. Components of Data Communication

A data communication system has five essential components:

1. **Sender:** The device that sends the data. (e.g., a computer, a phone).
2. **Receiver:** The device that receives the data. (e.g., a server, another computer).
3. **Message:** The actual data or information being communicated. (e.g., text, image, audio, video).
4. **Communication Media:** The physical path through which the message travels from sender to receiver. (e.g., cables, radio waves).
5. **Protocols:** A set of rules that govern the communication process. Both the sender and receiver must follow the same protocol for successful communication. (e.g., HTTP, TCP/IP).

3. Measuring Capacity of Communication Media

Bandwidth:

- **Definition:** The range of frequencies that a communication medium can carry. It is the difference between the highest and lowest frequencies.
- **Unit:** Hertz (Hz), Kilohertz (kHz), Megahertz (MHz), Gigahertz (GHz).
- **Analogy:** Think of it as the width of a highway. A wider highway (more bandwidth) can accommodate more cars (data) at the same time.

Data Transfer Rate:

- **Definition:** The amount of data that can be transferred from one point to another in a given amount of time. It is the actual speed of data transmission.
- **Unit:** Bits per second (bps), Kilobits per second (Kbps), Megabits per second (Mbps), Gigabits per second (Gbps).
- **Note:** Bandwidth is a *potential* capacity, while Data Transfer Rate is the *actual* speed achieved.

4. IP Address

> **Definition:** A unique numerical label assigned to every device connected to a computer network that uses the Internet Protocol for communication.

- **Purpose:** It serves two main functions: host or network interface identification and location addressing.
- **Example:** `192.168.1.1` (IPv4) or `2001:0db8:85a3::8a2e:0370:7334` (IPv6).

> **Analogy:** It's like a postal address for your computer on the network.

5. Switching Techniques

Switching is the method used to establish a connection between a sender and a receiver in a network.

Circuit Switching:

- **How it works:** A dedicated physical path (circuit) is established between the sender and receiver for the entire duration of the communication.

> **Analogy:** A traditional telephone call. Once you dial and connect, a dedicated line is reserved for you until you hang up.

Pros:

- Guaranteed data rate
- No delays

Cons:

- Inefficient use of resources (the line is idle if no data is being sent)
- Expensive

Packet Switching:

- **How it works:** The message is broken down into smaller units called **packets**. Each packet is sent independently across the network and may take different paths to reach the destination. The packets are reassembled at the destination.

> **Analogy:** Sending a book by tearing out its pages and mailing each page in a separate envelope. They might arrive out of order and via different postal routes, but they can be reassembled into the complete book.

Pros:

- Efficient use of network resources
- More resilient

Cons:

- Packets can be delayed or lost, requiring reassembly and retransmission

Difference between Circuit Switching and Packet Switching

Feature	Circuit Switching	Packet Switching
Connection Type	A dedicated physical path (circuit) is established before communication begins	No dedicated path; data is broken into packets sent independently
Path	Fixed path for the entire duration of communication	Each packet may take a different path to reach the destination
Resource Allocation	Resources are reserved for the entire session	Resources are shared among all users
Bandwidth Usage	Inefficient — bandwidth is wasted when no data is being sent	Efficient — bandwidth is utilized only when packets are sent
Delay	Minimal delay once the circuit is established (no routing delay)	Variable delay due to routing and queuing at each node
Order of Delivery	Data arrives in order (since path is fixed)	Packets may arrive out of order and need reassembly
Reliability	Reliable once connected; no packet loss	Less reliable; packets can be lost or delayed
Cost	Expensive (dedicated resources)	Cheaper (shared resources)
Setup Time	Requires call setup time before data transfer	No setup time required; data can be sent immediately

Feature	Circuit Switching	Packet Switching
Failure Impact	If any link in the circuit fails, the entire connection breaks	If one path fails, packets can be rerouted through another path
Example	Traditional telephone call (PSTN)	Internet communication (email, web browsing, VoIP)
Best For	Real-time voice communication where delay is unacceptable	Data communication where efficiency is more important than delay

> **Analogy:**
> **Circuit Switching:** A traditional telephone call — once you dial and connect, a dedicated line is reserved for you until you hang up.
> **Packet Switching:** Sending a book by tearing out its pages and mailing each page in a separate envelope — they might arrive out of order and via different routes, but can be reassembled into the complete book.

Part 3: Transmission Media

Transmission media is the physical channel that carries the data signal.

1. Wired Communication Media (Guided Media)

Twisted Pair Cable:

- **Description:** Consists of two insulated copper wires twisted around each other. The twisting helps to reduce electromagnetic interference (crosstalk).
- **Types:**
  - **Unshielded Twisted Pair (UTP):** Most common, used in Ethernet networks and telephones. (e.g., Cat5, Cat6 cables).
  - **Shielded Twisted Pair (STP):** Has a metal foil or braid around the pairs for extra protection from interference. More expensive and harder to install.

Pros:

- Cheap
- Easy to install

Cons:

- Low bandwidth
- Susceptible to interference
- Short range

Co-axial Cable:

- **Description:** Has a single copper conductor at its center, surrounded by an insulating layer, a braided metal shield, and an outer plastic jacket.
- **Use:** Used for cable TV and older Ethernet networks.

Pros:

- Higher bandwidth than twisted pair
- Better protection from interference

Cons:

- Thicker and less flexible than twisted pair
- More expensive

Fiber-optic Cable:

- **Description:** Uses thin strands of glass or plastic (fibers) to transmit data as pulses of light.
- **Use:** Backbone of the Internet, high-speed LANs, long-distance communication.

Pros:

- Extremely high bandwidth
- Very long range
- Immune to electromagnetic interference
- Very secure

Cons:

- Expensive
- Difficult to install and splice

2. Wireless Communication Media (Unguided Media)

Radio Waves:

- **Description:** Electromagnetic waves with frequencies from 3 kHz to 3 GHz. They can travel long distances and penetrate walls.
- **Use:** AM/FM radio, television, cordless phones, Wi-Fi, Bluetooth.

Pros:

- Can travel long distances
- Penetrate obstacles

Cons:

- Susceptible to interference from other radio sources

Micro Waves:

- **Description:** Electromagnetic waves with frequencies from 300 MHz to 300 GHz. They travel in straight lines and require a clear line of sight between the transmitter and receiver.
- **Use:** Satellite communication, microwave ovens, radar.

Pros:

- High bandwidth

Cons:

- Requires line-of-sight
- Affected by weather (rain fade)

Infrared Waves:

- **Description:** Electromagnetic waves with frequencies from 300 GHz to 400 THz. They are used for very short-range communication.
- **Use:** TV remote controls, wireless keyboards and mice, some short-range LANs.

Pros:

- Secure (cannot penetrate walls)
- Inexpensive

Cons:

- Very short range
- Requires line-of-sight

Part 4: Network Devices

Device	Description
Modem (MOdulator-DEModulator)	A device that converts digital signals from a computer into analog signals (modulation) for transmission over telephone lines, and converts incoming analog signals back into digital signals (demodulation).
Ethernet Card (NIC — Network Interface Card)	A hardware component installed in a computer that allows it to connect to a network. It provides the physical connection (e.g., for an Ethernet cable) and has a unique MAC address.
RJ45 (Registered Jack 45)	The standard connector used for Ethernet cables (like Cat5/Cat6). It looks like a slightly larger telephone connector.
Repeater	A device that regenerates and amplifies a weak or degraded signal over a long distance, allowing it to travel further without losing quality.

Device	Description
Hub	A basic networking device that connects multiple devices in a LAN. It is a **dumb** device that broadcasts any data it receives to all connected devices, leading to unnecessary traffic and collisions.
Switch	An intelligent networking device that connects multiple devices in a LAN. It learns the MAC addresses of connected devices and forwards data only to the intended recipient, making it more efficient than a hub.
Router	A device that connects different networks together (e.g., your home LAN to the Internet). It uses IP addresses to determine the best path for data packets to travel.
Gateway	A device that acts as a "gate" between two networks that use different protocols. It translates data from one protocol to another, allowing dissimilar networks to communicate.
Wi-Fi Card	A wireless network interface controller that allows a device to connect to a network using Wi-Fi (Wireless Fidelity).

MAC Address (Media Access Control Address)

MAC Address stands for **Media Access Control Address**. It is a unique 48-bit (6-byte) hardware address assigned to a Network Interface Card (NIC) by the manufacturer. It is also known as a **physical address** or **hardware address**.

Examples of MAC Addresses:

- `00:1A:2B:3C:4D:5E` — A typical Ethernet NIC
- `A4:5E:60:12:34:56` — A Wi-Fi adapter
- `00:50:56:C0:00:08` — A VMware virtual machine NIC (note: VMware uses its own OUI)
- `3C:5A:B4:9F:1D:22` — A Bluetooth adapter

MAC vs IP Address: A MAC address is a permanent physical address (like a fingerprint), while an IP address is a logical address that can change (like a postal address). MAC works at Layer 2, IP works at Layer 3.

Placement of a Repeater

Cable Type	Maximum Segment Length	Purpose of Repeater
Twisted Pair (UTP)	**100 metres**	Used to regenerate the signal and extend the network beyond 100 m

Remember: In standard Ethernet using twisted-pair cable, a cable segment can be up to **100 metres**. A repeater is used when the network needs to extend beyond this distance.

Difference between Hub and Switch, Router and Gateway

a) Hub vs Switch

Feature	Hub	Switch
Function	Connects multiple devices in a network	Connects multiple devices and forwards data to the intended device
Data Transmission	Sends data to all connected devices	Sends data only to the intended device
Efficiency	Less efficient	More efficient
Security	Low — all devices see all traffic	High — traffic is isolated
Cost	Cheaper	More expensive
Usage	Used in older or simple networks	Commonly used in LANs such as school and office networks

Example Scenario: In a school computer lab with 30 computers, if a **Hub** is used, when Computer A sends data to Computer B, all 30 computers receive the data. This wastes bandwidth and causes collisions. If a **Switch** is used, the data goes only to Computer B, making the network faster and more secure.

b) Router vs Gateway

Feature	Router	Gateway
Function	Connects different networks and routes data between them	Connects networks that may use different communication protocols
Main Purpose	Determines the appropriate path for data packets	Acts as a connection point between different networks or systems
Complexity	Less complex	More complex
Speed	Faster	Slower (due to protocol conversion)
Example	Connects a home or school LAN to the Internet	Connects networks using different protocols

> **Example Scenario (Router):** Your home Wi-Fi router connects your home LAN (using private IPs like 192.168.1.x) to the Internet (using public IPs). It routes packets between these two IP-based networks.
>
> **Example Scenario (Gateway):** A smart TV may use a gateway to communicate between its home network and a different network or service that uses a different communication protocol.

Part 5: Network Topologies and Network Types

1. Types of Networks (Based on Geographical Area)

- **PAN (Personal Area Network):** A network organized around an individual person. Range is typically up to 10 meters. (e.g., Bluetooth connection between a phone and a headset).
- **LAN (Local Area Network):** A network that connects devices within a small, limited area like a home, office, or school. (e.g., your home Wi-Fi network).
- **MAN (Metropolitan Area Network):** A network that spans a city or a large campus. It is larger than a LAN but smaller than a WAN. (e.g., a cable TV network in a city).
- **WAN (Wide Area Network):** A network that spans a large geographical area, such as a country, continent, or the entire world. The Internet is the largest WAN. (e.g., a bank's network connecting its branches across the country).

2. Range, Examples, Use Cases

Network Type	Full Form	Geographical Area Covered	Examples	Use Case
PAN	Personal Area Network	Very small area around a person (Up to 10 m)	Bluetooth headset, smartwatch, wireless keyboard, NFC payment	Connecting personal devices around one person
LAN	Local Area Network	Typically a building or a campus (Up to 1 km)	Home Wi-Fi, school computer lab, office network, college campus	Sharing files, printers, and internet within a small area
MAN	Metropolitan Area Network	A city or a metropolitan area or a large campus (10 km to 100 km)	Cable TV network, city-wide Wi-Fi, bank branches in a city, metro rail network	Connecting multiple LANs across a city
WAN	Wide Area Network	Large geographical area — countries or worldwide (more than 100 kms)	The Internet, national banking network, airline reservation system, mobile phone network	Connecting networks across countries and continents

More Examples with Use Cases:

- **PAN:** A person using Bluetooth headphones connected to their Phone via Bluetooth — use case: listening to music while walking.
- **LAN:** A family connecting their laptop, printer, and smart TV to a home Wi-Fi router — use case: sharing a printer and streaming movies.
- **MAN:** A city's traffic monitoring system where cameras across the city send data to a central control room — use case: traffic management.
- **WAN:** A multinational company connecting its offices in India, USA, and UK — use case: sharing company data and video conferencing across continents.

3. Networking Topologies

Topology refers to the physical or logical arrangement of devices in a network.

Bus Topology:

- **Layout:** All devices are connected to a single central cable called the “backbone” or “bus.”

Pros:

- Easy to install
- Requires less cable

Cons:

- If the backbone cable fails, the entire network goes down
- Difficult to troubleshoot

Star Topology:

- **Layout:** All devices are connected to a central device, such as a hub or a switch.

Pros:

- Easy to install and manage
- If one device or cable fails, only that device is affected
- Easy to add or remove devices

Cons:

- If the central hub/switch fails, the entire network goes down

Tree Topology:

- **Layout:** A hybrid topology that combines characteristics of Bus and Star topologies. It consists of groups of star-configured networks connected to a linear bus backbone cable.

Pros:

- Scalable
- Easy to manage in segments

Cons:

- If the backbone cable fails, the entire network is affected

4. Use Case for Topologies

Topology	Preferred Scenario	Why?
Bus	Small networks with few devices (e.g., 5–10 computers) where cost is a major concern and network traffic is low	Requires least cable, easy to install, cheap
Star	Office LANs, school computer labs, home networks where reliability and easy troubleshooting are important	If one device fails, others are unaffected; easy to add/remove devices; easy to diagnose faults
Tree	Large organizations, university campuses, multi-floor buildings where the network	Scalable; allows hierarchical grouping of departments; easy to

Topology	Preferred Scenario	Why?
	needs to be divided into segments	manage segments

Example Scenarios:

- **Bus:** A small cyber café with 6 computers on a tight budget.
- **Star:** A school computer lab with 40 computers connected to a central switch — if one computer's cable is faulty, only that computer is affected.
- **Tree:** A university with separate departments (Science, Arts, Commerce) each having its own star network, all connected to a central backbone.

5. Difference between Bus and Star Topology

Feature	Bus Topology	Star Topology
Layout	All devices connected to a single backbone cable	All devices connected to a central hub/switch
Cable Requirement	Less cable required	More cable required (each device needs its own cable)
Cost	Cheaper	More expensive (due to central device and extra cable)
Failure Impact	If backbone cable fails, **entire network** goes down	If central hub/switch fails, **entire network** goes down; if one cable fails, only that device is affected
Troubleshooting	Difficult — fault in backbone affects all	Easy — isolate the faulty cable/device
Scalability	Difficult to add new devices (signal degradation)	Easy to add new devices (just connect to hub/switch)
Data Collision	High chance of collisions	Low chance of collisions (especially with switch)
Performance	Degrades as more devices are added	Consistent performance
Best For	Small, temporary networks	Office, school, and home networks

Part 6: Network Protocols

A protocol is a set of rules for communication.

Protocol	Full Form	Description
HTTP	HyperText Transfer Protocol	The foundation of data communication for the World Wide Web. It is used to transfer web pages from a web server to a web browser.
FTP	File Transfer Protocol	Used for transferring files from one computer to another over a network (e.g., from a client to a server).
PPP	Point-to-Point Protocol	A data link protocol used to establish a direct connection between two nodes. It is commonly used for dial-up and DSL Internet connections.
SMTP	Simple Mail Transfer Protocol	The standard protocol for **sending** email messages from a client to a server or between servers.
TCP/IP	Transmission Control Protocol / Internet Protocol	The fundamental protocol suite of the Internet. **TCP** breaks data into packets and ensures they arrive correctly. **IP** handles the addressing and routing of these packets.
POP3	Post Office Protocol version 3	A protocol used by email clients to **retrieve** (download) emails from a mail server. The emails are typically deleted from the server after download.
HTTPS	HyperText Transfer Protocol Secure	A secure version of HTTP. It uses encryption (SSL/TLS) to protect the data being transferred between the browser and the web server. Used for online banking, shopping, etc.
TELNET	Terminal Network	A protocol used to remotely log in to another computer on a network. It is a text-based, unencrypted protocol.
VoIP	Voice over Internet Protocol	A technology that allows you to make voice calls using a broadband Internet connection instead of a regular phone line. (e.g., Skype, WhatsApp calls).

Examples for the Protocols

Protocol	Full Form	Real-World Examples
HTTP	HyperText Transfer Protocol	Opening `http://www.example.com` in a browser; loading a news website

Protocol	Full Form	Real-World Examples
FTP	File Transfer Protocol	Uploading a website to a web server using FileZilla; downloading a file from an FTP server
PPP	Point-to-Point Protocol	Connecting to the Internet via a DSL modem; dial-up Internet connection
SMTP	Simple Mail Transfer Protocol	Sending an email from Gmail to Yahoo; sending an email from Outlook
TCP/IP	Transmission Control Protocol / Internet Protocol	Every Internet communication — browsing, email, video streaming, online gaming
POP3	Post Office Protocol version 3	Downloading emails from a mail server to Microsoft Outlook; configuring Gmail in a desktop email client
HTTPS	HyperText Transfer Protocol Secure	Online banking (`https://www.onlinesbi.com`); shopping on Amazon; logging into Gmail
TELNET	Terminal Network	Remote login to a server; configuring a router via command line; accessing a remote UNIX/Linux machine
VoIP	Voice over Internet Protocol	WhatsApp voice call; Skype call; Zoom meeting; Google Meet

Part 7: Introduction to Web Services

Term	Description
WWW (World Wide Web)	An information system where documents and other web resources are identified by URLs, interlinked by hypertext links, and can be accessed via the Internet. It is a service that runs *on* the Internet.
Hyper Text Markup Language (HTML)	The standard markup language for creating web pages and web applications. It describes the structure of a web page.
Extensible Markup Language (XML)	A markup language designed to store and transport data. Unlike HTML, XML has no predefined tags; you create your own tags to describe the data.

Term	Description
Domain Names	A human-readable name that identifies a website (e.g., `google.com`). It is mapped to a numeric IP address by the Domain Name System (DNS).
URL (Uniform Resource Locator)	The complete address of a resource on the web. It consists of the protocol, domain name, and path to the specific file. (e.g., `https://www.example.com/index.html`).
Website	A collection of related web pages and other digital assets that are typically identified with a common domain name and published on at least one web server.
Web Browser	A software application used to access and view websites. (e.g., Google Chrome, Mozilla Firefox, Safari).
Web Servers	A computer system that hosts websites and delivers web pages to clients (web browsers) upon request. (e.g., Apache, Nginx).
Web Hosting	A service that provides the storage space and technology needed for a website to be accessible on the Internet. A web hosting company rents out server space to individuals and organizations.

Examples

Web Service	Examples	Use Case
WWW (World Wide Web)	Google, Wikipedia, YouTube, Amazon	Accessing information, watching videos, online shopping
HTML	`<h1>Welcome</h1>`, `<p>This is a paragraph</p>`, `<img src="photo.jpg">`	Creating the structure and content of web pages
XML	`<student><name>Rahul</name> <roll>12</roll></student>`	Storing and transporting data between applications (e.g., RSS feeds, configuration files)
Domain Names	`google.com`, `cbse.gov.in`, `wikipedia.org`, `amazon.in`	Human-readable addresses for websites
URL	`https://www.google.com/search?q=class12`, `https://www.youtube.com/watch?v=abc123`	Complete address to access a specific web resource

Web Service	Examples	Use Case
Website	`www.cbse.gov.in` (CBSE official website), `www.flipkart.com`	A collection of related web pages under one domain
Web Browser	Google Chrome, Mozilla Firefox, Microsoft Edge, Safari, Opera	Software used to access and view websites
Web Servers	Apache HTTP Server, Nginx, Microsoft IIS, LiteSpeed	Hosting websites and serving web pages to browsers
Web Hosting	GoDaddy, Bluehost, HostGator, AWS (Amazon Web Services), GitHub Pages	Renting server space to make a website accessible on the Internet

Complete Example — How Web Services Work Together:

1. A student types `https://www.cbse.gov.in` in **Google Chrome** (Web Browser).
2. The **URL** is resolved via **DNS** to an IP address.
3. The request goes to the **Web Server** (e.g., Apache) hosting the CBSE **Website**.
4. The server sends back **HTML** pages, which the browser renders.
5. The website is made available through **Web Hosting** services.
6. **HTTPS** ensures the connection is secure.